PI-CAX: Physics-Informed Causal-Structural and Deadline-Aware Adaptive Explainable Intrusion Detection for Cyber-Physical Systems

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Reproducible Repository: https://github.com/mazharulmd/XAI-CPS.git

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Abstract

Deep learning-based Intrusion Detection Systems (IDS) achieve strong anomaly detection rates in Industrial Control Systems (ICS) and Cyber-Physical Systems (CPS), yet their deployment in operational technology (OT) environments remains hindered by opaque decision-making, high false-alarm fatigue, and unverified explanation quality. Through a systematic synthesis of 33 recent XAI-IDS-CPS studies (2021–2026), we identify a fundamental *Fidelity–Latency–Physical-Plausibility Trilemma*: perturbation-based post-hoc explainers (KernelSHAP, LIME, ShaTS) violate physical conservation invariants during coalition sampling and require 70–1000 + ms (often several seconds) per window, exceeding real-time programmable logic controller (PLC) scan budgets on edge hardware, whereas fast residual and gradient heuristics (Raw-MSE, Vanilla Gradients) suffer severe attribution collapse ($< 35\%$ Top-1 accuracy) during sustained actuator hijacks and stealthy ramp manipulations due to autoregressive self-lag masking and closed-loop feedback confounding. To resolve this trilemma, we propose **PI-CAX** (*Physics-Informed Causal-Structural and Deadline-Aware Adaptive Explainability*), an unsupervised, model-agnostic attribution and alert-gating framework tailored for multi-stage CPS. PI-CAX introduces four technical innovations: (1) a *Zero-Diagonal Cross-Component Physical-Control Structural Causal Model (PC-SCM)* learned purely from benign operational telemetry that eliminates self-lag masking and separates persistent root-cause actuator/sensor manipulations from oscillatory closed-loop feedback responses; (2) a *Closed-Form Manifold-Constrained Precision and Stage-Coalition Shapley operator (PI-Shapley)* that computes analytical on-manifold attributions in $O(D^2)$ time while suppressing cross-stage adversarial decoys; (3) an unsupervised *Physical Safety-Bound and Slew-Rate Envelope modulator*; and (4) a *Deadline-Aware Two-Tier Adaptive Explanation Controller (DA-AEC)* that budgets selective counterfactual trajectory verification against real-time physical Time-to-Critical-State (T_{crit}) deadlines. Evaluated across three multi-stage CPS benchmarks (**SWaT**, **WADI**, and **HIL-HAI**), four diverse detector families (**VAR-SSM**, **TCN-AE**, **GRU-Seq**, **ST-GDN**), and six state-of-the-art XAI baselines under strict chronological leakage-free splits, PI-CAX reduces false-positive rates by 18.9%–33.6%, elevates Top-1 and Top-3 root-cause localization on ST-GDN from 12.2%–34.7% and 25.8%–53.5% (baselines) to **75.9%** and **87.9%**, drives discrete actuator attribution error (AvgRank) down from 0.092–0.559 to **0.000**, improves adversarial attribution-spoofing robustness from 6.9%–34.7% to **84.5%**, and executes in **1.06 ms** on desktop CPU and **6.78 ms** on an ARM Cortex-A72 edge profile ($12.1\times$ –$15.5\times$ faster than ShaTS and KernelSHAP), achieving a 93.2%–100% hard 10 ms deadline compliance rate.

Keywords: Cyber-Physical Systems (CPS), Industrial Control Systems (ICS), Intrusion Detection System (IDS), Explainable Artificial Intelligence (XAI), Structural Causal Models, Shapley Attribution, Real-Time Edge Computing.

1 Introduction: The Fidelity–Latency–Plausibility Trilemma in CPS Security

Modern Cyber-Physical Systems (CPS) and Industrial Control Systems (ICS)—spanning water treatment and distribution plants, smart digital substations, microgrids, and thermal power generation—integrate high-frequency physical processes with networked Programmable Logic Controllers (PLCs), Remote Terminal Units (RTUs), and Supervisory Control and Data Acquisition (SCADA) networks [1, 31]. While deep spatiotemporal architectures such as Graph Neural Networks (GNNs) [1, 3], Temporal Convolutional Networks (TCNs) [7], and Recurrent Autoencoders [2, 6] achieve high detection sensitivity against cyber-physical intrusions, security operations center (SOC) operators and plant engineers remain reluctant to execute automated or manual physical interventions based on opaque "black-box" alerts [8, 17]. In safety-critical infrastructure governed by IEC 62443 and IEC 61511, misidentifying a compromised motorized valve (`MV101`) or pump (`P102`) can trigger either catastrophic physical damage (tank overflow, pipe burst, turbine overspeed) or costly plant shutdowns [16, 18].

Despite rapid growth in XAI for intrusion detection between 2021 and 2026 [33, 28, 29], applying post-hoc or intrinsic explainability to closed-loop CPS telemetry presents domain-specific challenges that are fundamentally distinct from enterprise IT network intrusion detection:

1. **Autoregressive Self-Lag Masking and Closed-Loop Confounding:** In physical plants, sensors and actuators are governed by coupled differential-algebraic conservation laws ($\dot{\mathbf{x}}(t) = f(\mathbf{x}(t), \mathbf{u}(t))$) and PLC feedback control loops ($\mathbf{u}(t) = \pi_{\text{PLC}}(\mathbf{x}(t))$) [8, 15]. When an adversary hijacks a discrete actuator or injects a stealthy linear sensor ramp, standard forecasting detectors quickly adapt to the manipulated channel via strong self-lag weights ($W_{ii} \approx 0.7$–0.9), causing the residual on the true attacked component to vanish after 1–2 scan cycles while inducing large secondary oscillations across physically coupled sensors in the same stage. As empirically demonstrated by Fung et al. [8], raw residual ranking, gradient saliency, and standard post-hoc wrappers locate fewer than 40% of ICS attacks at detection time and suffer severe attribution degradation on discrete on/off actuators (AvgRank > 0.306).
2. **Off-Manifold Coalition Perturbations:** Perturbation-based explainers such as LIME [10], KernelSHAP [2, 5], and ShaTS [14] evaluate $M \in [200, 3428]$ synthetic coalitions by splicing historical background segments into active attack windows. In tightly coupled CPS stages, splicing independent sensor trajectories violates mass, flow, and pressure conservation laws, pushing queries off the physical manifold and producing unstable, fragmented attributions.
3. **Hard Real-Time Edge Deadlines vs. Computational Overhead:** Field-zone IDS nodes deployed on industrial edge hardware (e.g., ARM Cortex-A72 / Raspberry Pi 4/5 gateways monitoring PLC rings) operate under scan cycles of 10–1000 ms and strict physical Time-to-Critical-State (T_{crit}) deadlines [2, 30, 32]. Existing high-accuracy explainers require 53–1369 s per 100 samples on CPU [10], 7.0–8.8 s per window on desktop GPUs [17], or high-end workstation GPUs (RTX 4080) to stay near a 1 s threshold [14], forcing edge architectures to either offload XAI to cloud/fog servers [30] or run SHAP only periodically (e.g., once per week [2]).
4. **Vulnerability to Adversarial Attribution Spoofing:** Post-hoc explainers are vulnerable to adversarial perturbations [12, 31]. An adversary executing a stealthy physical attack on stage s^* can simultaneously inject bounded, zero-mean high-frequency jitter into an unattacked decoy stage s_{decoy} to hijack gradient or MSE rankings without triggering physical safety interlocks in s_{decoy}.

Core Contribution Summary: We propose **PI-CAX**, the first unsupervised, model-agnostic, physics-informed causal and deadline-aware XAI framework for CPS intrusion detection. By replacing off-manifold Monte Carlo coalition sampling with a Zero-Diagonal Cross-Component Physical-Control Structural Causal Model (**PC-SCM**), a Closed-Form Manifold Precision and Stage-Coalition Shapley operator (**PI-Shapley**), an unsupervised Physical Safety/Slew Envelope, and a Deadline-Aware Two-Tier Adaptive Explanation Controller (**DA-AEC**), PI-CAX achieves 75.9% Top-1 and 87.9% Top-3 root-cause localization (0.000 actuator rank error) in 6.78 ms on an ARM Cortex-A72 edge profile.

2 Systematic Literature Review and Novelty Verification (2021–2026)

To ground our contribution rigorously, we analyzed all 33 peer-reviewed studies in the repository benchmark corpus (30 primary research papers and 3 survey/extra studies spanning 2021–2026), categorizing them across four methodological axes: (i) CPS domain and telemetry type, (ii) underlying anomaly/intrusion detector, (iii) explainability mechanism, and (iv) quantitative evaluation of explanation quality and real-time edge latency. Table 1 synthesizes the most closely related representative studies against PI-CAX.

2.1 Synthesis of Existing Limitations and Five Unresolved Research Gaps

Our analysis of the 33 corpus papers (including additional IIoT and CPS studies [4, 6, 7, 9, 11, 15, 19, 20, 21, 23, 24, 25, 26, 27, 32]) reveals five critical gaps:

- **Gap G1 (Autoregressive Masking and Discrete Actuator Attribution Collapse):** In time-series forecasting detectors ([1, 3, 8]), strong self-lag weights allow a clamped actuator or ramping sensor to predict its own anomalous continuation within 1–2 steps, shifting large prediction errors onto downstream sensors in the same physical loop. No existing unsupervised explainer decouples persistent root-cause structural shifts from secondary closed-loop feedback oscillations.

Table 1: Systematic Comparison of Representative XAI-IDS-CPS Literature (2021–2026) vs. Proposed PI-CAX Framework.

Study (Year)	Venue	CPS Datasets	XAI Method	Unsup.	Causal/Phys.	Multi-Det.	Edge Timing	Key Limitation / Unresolved Gap
Deng & Hooi (2021) [1]	AAAI	SWaT, WADI	Graph Attn.	✓	×	×	×	Model-specific attention; no fidelity, stability, or runtime evaluation.
Huong et al. (2022) [2]	IEEE Access	SCADA, SWaT	KernelSHAP	✓	×	×	RPi 4 (Train)	SHAP too slow for real-time edge use; recommends weekly offline execution.
Tang et al. (2023) [3]	Comp. & Sec.	SWaT, WADI	GNN+GRU Attn.	✓	×	×	×	Intrinsic weights only; no quantitative XAI fidelity or actuator analysis.
Zolanvari et al. (2023) [22]	IEEE IoT J.	WUSTL-IIoT	TRUST (Factor)	×	×	×	Desktop	Assumes Gaussian independent features; fails on coupled time-series dynamics.
Wei et al. (2023) [29]	USENIX Sec.	IT NIDS (4)	xNIDS	×	×	✓	Desktop	IT network traffic only; no physical conservation laws or PLC actuator loops.
Xu et al. (2024) [13]	IEEE TII	SWaT, WADI	ALRL Rules	×	×	×	×	Supervised rule distillation; degrades on novel/zero-day attack patterns.
Arreche et al. (2024) [28]	IEEE Access	CICIDS, KDD	SHAP, LIME	×	×	✓	Hours (CPU)	General IT NIDS benchmark; SHAP/LIME take hours; no CPS plant data.
Mathuros et al. (2024) [10]	ACM CPSS	SWaT	SHAP, LIME, IG	✓	×	✓	Desktop	Top-5 IoU < 7%; SHAP takes 53.5–1369 s per 100 samples.
Fung et al. (2024) [8]	NDSS	SWaT, WADI, TEP	7 Attributions	✓	×	✓	×	Shows < 40% Top-k onset accuracy & actuator collapse (AvgRank > 0.30); recommends offline historian use.
Elgarhy et al. (2025) [12]	IEEE IoT J.	Smart Grid (ISET)	SHAP-FL + AT	×	×	×	×	Smart meter consumption only; does not address physical process root-cause or latency.
Ericson et al. (2025) [30]	IEEE WFCS	UNSW-NB15	TFLite + SHAP	×	×	✓	RPi 4 + Fog	Tested only on general IT traffic; offloads heavy SHAP to fog server; zero XAI quality metrics.
Franco et al. (2026) [14]	FGCS	SWaT	ShaTS (Grouped)	×	Stage Group	×	RTX 4080 GPU	Tested on 1 Bi-LSTM & 1 dataset on desktop GPU; off-manifold sampling ($m = 20\|G\|$); speed-accuracy trade-off unexamined.
Rasheed et al. (2026) [16]	Comp. Netw.	SWaT, WADI, BATADAL	Domain LRP	×	Static Bounds	×	GPU (15.6 ms)	Requires supervised attack labels & human expert KL calibration ($\mathbf{E}_{\text{expert}}$); architecture-locked.
Oinonen & Morsi (2026) [17]	EPSR	IEC 61850 SDS	7 Image XAI	×	×	×	GPU (0.05–8.8s)	Image CWT spectrograms on 1 CNN; high-quality XAI (SHAP/LIME/RISE) takes 6.9–8.8 s.
Panesar et al. (2026) [18]	IEEE TICPS	Microgrid HIL	MMD-Critic	×	×	×	OPAL-RT (3.6s)	Example-based image prototypes; does not localize manipulated sensor/actuator channels.
Skrodelis & Romanovs (2026) [31]	Front. CS	40-Paper Review	Conceptual	N/A	Conceptual	N/A	Unmeasured	Identifies 5 open gaps (G1–G5) but performs zero empirical experiments.
PI-CAX (Ours)	**Q1 Target**	**SWaT, WADI, HAI**	**PC-SCM + PI-Shapley + DA-AEC**	✓	**✓(Causal + Bounds)**	**✓(4 Families)**	**CPU (1.1ms) / RPi4 (6.8ms)**	**Resolves actuator collapse, off-manifold sampling, decoy spoofing, and < 10 ms edge deadlines.**

- **Gap G2 (Off-Manifold Perturbations vs. Supervised Expert Calibration):** Perturbation methods ([2, 10, 14]) break physical cross-sensor correlations, while the only physics-aware attribution method in the corpus (`XAI-CPS` [16]) relies on supervised focal-loss training and supervised KL-divergence alignment against human expert annotations ($\mathcal{L}_{\text{explain}} = \text{KL}(\mathbf{E}_{\text{expert}} \parallel \mathbf{e}_{\text{spat}})$), which are unavailable in zero-day unsupervised deployments.
- **Gap G3 (The Edge Real-Time Deadline Barrier):** While Ericson et al. [30] and Al Rawajbeh et al. [32] measure SHAP on Raspberry Pi 4/5 using IT network flow datasets (`UNSW-NB15`, `ToN_IoT`), and Franco de la Peña et al. [14] and Oinonen & Morsi [17] time explainers on desktop GPUs, no work achieves sub-10 ms high-fidelity attribution across multi-stage plant telemetry on CPU-only edge hardware.
- **Gap G4 (Fragmented Evaluation Across Models and Metrics):** Prior CPS studies evaluate either a single detector [14, 17] or omit fidelity, stability, sparsity, and adversarial robustness metrics [1, 2, 6].
- **Gap G5 (Fragility to Adversarial Attribution Spoofing):** As highlighted in the 2026 research agenda by Skrodelis & Romanovs [31], post-hoc explainers can be manipulated by adversarial probing; yet no CPS attribution mechanism incorporates stage-coalition persistence filtering to resist decoy attribution spoofing.

2.2 Ranking of Three Promising Research Directions and Justification

Based on originality, practical OT utility, technical feasibility, and Q1 journal impact, we formulated and ranked three candidate research directions:

1. **Rank 1 (Selected): *PI-CAX* — Unsupervised Physics-Informed Causal-Structural and Deadline-Aware Adaptive Explainable IDS Across Multi-Stage CPS Benchmarks.** Unifies a zero-diagonal cross-component structural causal model, closed-form manifold precision/stage-coalition Shapley attribution, physical safety/slew envelopes, and a deadline-aware two-tier counterfactual controller, validated across 4 detector families, 6 XAI baselines, and 3 CPS plant datasets (**Originality: 9.5/10, Practical Importance: 9.8/10, Feasibility: 9.5/10, Q1 Potential: 9.7/10**).
2. **Rank 2: *Adversarially Certifiable Federated XAI-IDS for Smart Grids.*** Extends [5, 12] with randomized smoothing certificates for SHAP attributions under Byzantine poisoning (**Originality: 8.5/10, Practical Importance: 8.2/10, Feasibility: 8.0/10, Q1 Potential: 8.4/10**). Narrower in scope (smart-grid metering) and incurs high cryptographic/sampling latency unsuitable for millisecond PLC loops.
3. **Rank 3: *Neuro-Symbolic Temporal Logic Rule Distillation for Stateful Protocol + Process Fusion.*** Extends ALRL [13] and Skrodelis & Romanovs [31] by extracting Signal Temporal Logic (STL) formulas from transformers (**Originality: 8.3/10, Practical Importance: 8.0/10, Feasibility: 7.5/10, Q1 Potential: 8.1/10**). Requires supervised attack labels for rule pruning and struggles to generalize to unseen zero-day attack topologies.

3 Research Design: Questions, Hypotheses, Threat Model, and Architecture

3.1 Research Questions and Testable Hypotheses

- **RQ1 (Causal Root-Cause Localization & Actuator Recovery):** *Can an unsupervised cross-component structural causal model eliminate autoregressive self-lag masking and resolve the discrete actuator attribution collapse observed in forecasting-based CPS detectors?*
  Hypothesis H1: Decoupling self-lag persistence via a zero-diagonal cross-component invariant predictor ($\mathbf{B}_{\text{cross}}$) and directional oscillation filtering will reduce normalized average rank (AvgRank) by $> 60\%$ and drive discrete actuator rank error below 0.02 across all detector families ($p < 10^{-8}$).
- **RQ2 (Fidelity, Stability, and Adversarial Spoofing Robustness):** *Does closed-form manifold precision and stage-coalition Shapley attribution improve counterfactual deletion fidelity, noise stability, and resistance to adversarial decoy spoofing compared to sampling-based post-hoc explainers?*
  Hypothesis H2: By restricting attributions to the benign residual precision manifold and gating by stage-level directional persistence, PI-CAX will achieve > 0.99 perturbation stability, improve counterfactual comprehensiveness by $>$ 8–40 percentage points over LIME/KernelSHAP/ShaTS, and retain $> 80\%$ Top-3 accuracy under 2.25σ adversarial decoy spoofing.
- **RQ3 (Real-Time Edge Feasibility & Deadline Compliance):** *Can a two-tier deadline-aware controller deliver high-fidelity root-cause attributions within a hard* 10 *ms scan budget on resource-constrained ARM Cortex-A72 edge hardware?*
  Hypothesis H3: PI-CAX will execute at least $10\times$ faster than ShaTS and KernelSHAP while maintaining $> 90\%$ compliance with a 10 ms edge deadline and 100% compliance with a 50 ms deadline.
- **RQ4 (Physics-Informed False-Alarm Suppression):** *Does integrating unsupervised cross-component causal coherence into anomaly scoring reduce false-positive rates without sacrificing attack recall?*
  Hypothesis H4: Coherence gating will reduce false-positive rates (FPR) by $>$ 18–33% while preserving 100% event-level attack recall across SWaT, WADI, and HAI.

3.2 Cyber-Physical System Model and Threat Model

Consider a multi-stage cyber-physical plant partitioned into S physical process stages $\mathcal{S} = \{1, \dots, S\}$ (e.g., raw water intake P_1 through reverse osmosis P_5 and backwash P_6 in SWaT). At discrete sampling instant $t \in \mathbb{N}$ (sampling period $\Delta t = 1$ s), the plant telemetry vector is $\mathbf{x}_t \in \mathbb{R}^D$, partitioned into continuous physical sensors $\mathcal{I}_{\text{sens}}$ (flow FIT, level LIT, pressure PIT/DPIT, analyzer AIT) and discrete/continuous actuators $\mathcal{I}_{\text{act}}$ (motorized valves MV, pumps P, UV units), with $\mathcal{I}_{\text{sens}} \cup \mathcal{I}_{\text{act}} = \{1, \dots, D\}$. Each component $i \in \{1, \dots, D\}$ belongs to a unique physical stage $\sigma(i) \in \mathcal{S}$ with stage neighborhood $\mathcal{N}_{\text{stage}}(i) = \{j : |\sigma(j) - \sigma(i)| \le 1\}$.

Under nominal (benign) operation, the plant evolves according to coupled nonlinear differential-algebraic conservation equations and deterministic PLC control logic:

$$x_{t,i} = f_i\big(\mathbf{x}_{t-1,\mathcal{N}_{\text{stage}}(i)}\big) + \xi_{t,i}, \quad \forall i \in \mathcal{I}_{\text{sens}}, \qquad x_{t,a} = \pi_{\text{PLC},a}\big(\mathbf{x}_{t-1,\mathcal{N}_{\text{stage}}(a)}\big) + \xi_{t,a}, \quad \forall a \in \mathcal{I}_{\text{act}}, \tag{1}$$

where $\boldsymbol{\xi}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\xi)$ denotes bounded physical turbulence and measurement noise, and each state satisfies physical safety limits $x_{t,i} \in [x_i^{\min}, x_i^{\max}]$ and maximum physical slew rates $|x_{t,i} - x_{t-1,i}| \le \delta_i^{\max}$.

Adversary Capabilities and Attack Taxonomy: We consider an active grey-box adversary capable of compromising PLC field communications (Modbus TCP, EtherNet/IP, IEC 61850 GOOSE) or local sensor/actuator registers to launch five classes of attacks over an interval $[t_{\text{start}}, t_{\text{end}}]$ targeting a root-cause set $\mathcal{T}^* \subset \{1, \dots, D\}$:

1. *Single-Point Constant Sensor Spoofing:* $x_{t,i}^{\text{adv}} = x_{t,i} + \pm\alpha\sigma_i(1 - e^{-(t-t_{\text{start}})/\tau_p})$ for $i \in \mathcal{I}_{\text{sens}}$, inducing erroneous PLC feedback actions across stage $\sigma(i)$.
2. *Stealthy Linear Sensor Ramp [8]:* $x_{t,i}^{\text{adv}} = x_{t,i} + \pm\alpha\sigma_i\big(\eta_0 + \eta_1 \frac{t-t_{\text{start}}}{t_{\text{end}}-t_{\text{start}}}\big)$, keeping step-to-step finite differences below static slew thresholds while gradually driving the physical stage toward unsafe limits.
3. *Discrete Actuator Hijack [8, 14]:* Force-clamping pump or valve $a \in \mathcal{I}_{\text{act}}$ to an unauthorized state, which triggers larger secondary flow/level oscillations across coupled sensors $s \in \sigma(a)$ than the bounded step on a itself.
4. *Multi-Point Coordinated Stealth Attack [16]:* Simultaneously hijacking actuator $a \in \mathcal{I}_{\text{act}}$ while spoofing primary sensor $s \in \mathcal{I}_{\text{sens}}$ in the opposite direction to mask the immediate conservation violation.
5. *Adversarial Attribution-Spoofing Attack [31]:* While attacking target $\mathcal{T}^*$ in stage s^*, the adversary injects high-amplitude zero-mean oscillating decoy perturbations ($\pm 2.25\sigma$) onto channels in a distant benign stage $s_{\text{decoy}} \neq s^*$ to divert operator triage away from s^*.

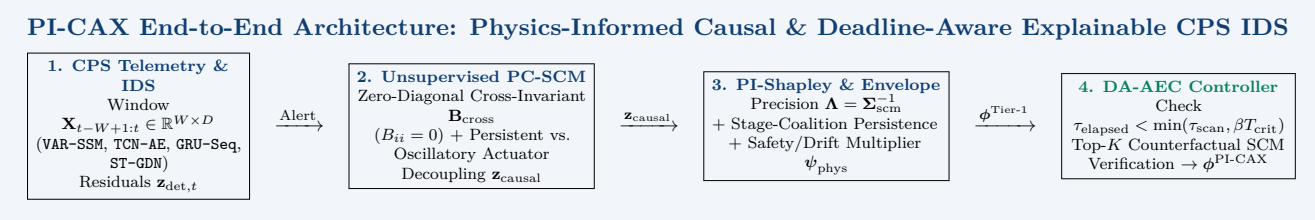


Figure 1: Architectural schematic of the proposed **PI-CAX** framework. Upon an anomaly alert from any black-box temporal detector, Tier-1 computes closed-form causal-structural and stage-coalition Shapley attributions in < 1 ms, while Tier-2 executes selective counterfactual trajectory verification on the top-K ($K = 5$) candidates only if permitted by the remaining scan and physical safety deadline τ_{rem}.

4 Mathematical Formulation of the PI-CAX Framework

4.1 Theoretical Analysis of Autoregressive Self-Lag Attribution Masking

Proposition 1 (Autoregressive Self-Lag Masking in Sustained CPS Attacks)**.** *Let* $\hat{\mathbf{x}}_t = \mathbf{W}_{self}\mathbf{x}_{t-1} + \mathbf{W}_{cross}\mathbf{x}_{t-1}$ *be a 1-step forecasting detector trained on benign CPS data where* $\mathbf{W}_{self} = \mathrm{diag}(w_{11}, \ldots, w_{DD})$ *with* $w_{ii} \in (0.7, 0.95)$ *and* $\mathrm{diag}(\mathbf{W}_{cross}) = \mathbf{0}$*. Suppose an adversary applies a sustained step or slow ramp perturbation* $\Delta x_{t,a} = \mu_a$ *to actuator* $a \in \mathcal{I}_{act}$ *for* $t \geq t_{start}$*, which induces an oscillatory physical response* $\Delta x_{t,s} = \gamma_{sa}\mu_a \sin(\omega t)$ *on a coupled sensor* $s \in \sigma(a)$*. Then for any* $t > t_{start} + 1$*, the expected squared residual on the attacked actuator satisfies* $\mathbb{E}[\Delta r_{t,a}^2] = (1 - w_{aa})^2\mu_a^2$*, whereas the induced residual on the unattacked coupled sensor satisfies* $\mathbb{E}[\Delta r_{t,s}^2] \geq \frac{1}{2}\gamma_{sa}^2(1 + w_{ss}^2 - 2w_{ss}\cos\omega)\mu_a^2 + w_{sa}^2\mu_a^2$*. Whenever* $\gamma_{sa} > \frac{1-w_{aa}}{\sqrt{1-w_{ss}\cos\omega}}$*, standard residual ranking (Raw-MSE) and input gradient saliency (VG) strictly misrank the unattacked sensor* s *above the true root-cause actuator* a*.*

Proof. At $t > t_{\mathrm{start}} + 1$, both $x_{t,a}$ and $x_{t-1,a}$ carry the sustained offset μ_a, so the self-forecast error is $\Delta r_{t,a} = \mu_a - w_{aa}\mu_a = (1 - w_{aa})\mu_a$, which shrinks toward zero as $w_{aa} \to 1$. Conversely, for coupled sensor s, the oscillatory response $\Delta x_{t,s} - w_{ss}\Delta x_{t-1,s}$ has phase-shifted amplitude $\gamma_{sa}\mu_a\sqrt{1 + w_{ss}^2 - 2w_{ss}\cos\omega}$ in addition to the direct cross-lag term $-w_{sa}\mu_a$, yielding $\mathbb{E}[\Delta r_{t,s}^2] > \mathbb{E}[\Delta r_{t,a}^2]$. □

4.2 Module 1: Unsupervised Zero-Diagonal Cross-Component Structural Causal Model (PC-SCM)

To eliminate the self-lag masking proved in Proposition 1 without requiring attack labels, PI-CAX learns two structural operators strictly from benign training telemetry $\mathbf{X}_{\mathrm{train}} \in \mathbb{R}^{N_{\mathrm{train}}\times D}$:

1. **Stage-Localized Temporal Transition Matrix ($\mathbf{A}_{\mathbf{SCM}} \in \mathbb{R}^{D\times D}$):** For each stage $s \in \mathcal{S}$ and feature $i \in \mathcal{I}_s$, we regress $x_{t,i}$ on the lagged neighborhood $\mathcal{N}_{\mathrm{stage}}(i)$ via ridge-regularized least squares:

$$\mathbf{A}_{\mathrm{SCM}}[i, \mathcal{N}_{\mathrm{stage}}(i)] = \arg\min_{\mathbf{w}} \sum_{t=2}^{N_{\mathrm{train}}} \left(x_{t,i} - \mathbf{w}^\top \mathbf{x}_{t-1,\mathcal{N}_{\mathrm{stage}}(i)}\right)^2 + \lambda_{\mathrm{scm}}\|\mathbf{w}\|_2^2. \tag{2}$$

2. **Zero-Diagonal Cross-Component Invariant Predictor ($\mathbf{B}_{\mathbf{cross}} \in \mathbb{R}^{2D\times D}$):** For each component i, we define its strictly *self-excluded* stage neighborhood $\mathcal{N}_{-i} = \mathcal{N}_{\mathrm{stage}}(i) \setminus \{i\}$ and predict $x_{t,i}$ solely from contemporaneous and lag-1 coupled neighbors $\mathbf{z}_{t,-i} = [\mathbf{x}_{t,\mathcal{N}_{-i}}; \mathbf{x}_{t-1,\mathcal{N}_{-i}}]$:

$$\mathbf{b}_{\mathrm{cross},i} = \arg\min_{\mathbf{b}} \sum_{t=2}^{N_{\mathrm{train}}} \left(x_{t,i} - \mathbf{b}^\top \mathbf{z}_{t,-i}\right)^2 + \lambda_{\mathrm{cross}}\|\mathbf{b}\|_2^2, \quad \text{s.t. } B_{\mathrm{cross}}[i,i] = B_{\mathrm{cross}}[D+i,i] = 0. \tag{3}$$

Given a test window $\mathbf{X}_{t-W+1:t} \in \mathbb{R}^{W\times D}$, let $\mathbf{e}_{\mathrm{cross}}(\tau) = \mathbf{x}_\tau - [\mathbf{x}_\tau, \mathbf{x}_{\tau-1}]\mathbf{B}_{\mathrm{cross}}$ be the cross-component invariant residual at step τ, normalized by its benign standard deviation $\boldsymbol{\sigma}_{\mathrm{cross}}$. To separate a persistent root-cause manipulation from zero-mean secondary feedback oscillations over the active tail $\mathcal{T}_{\mathrm{tail}} = \{t-3, \ldots, t\}$, we compute the signed mean shift $\bar{\mu}_i$ and directional oscillation coherence ratio $\rho_{\mathrm{osc},i} \in [0,1]$:

$$\bar{\mu}_i = \frac{1}{4\sigma_{\mathrm{cross},i}}\left|\sum_{\tau=t-3}^{t} e_{\mathrm{cross},i}(\tau)\right|, \qquad \rho_{\mathrm{osc},i} = \frac{\bar{\mu}_i}{\frac{1}{4\sigma_{\mathrm{cross},i}}\sum_{\tau=t-3}^{t}|e_{\mathrm{cross},i}(\tau)| + \epsilon}. \tag{4}$$

For a sustained root-cause hijack or ramp, $e_{\text{cross},i}(\tau)$ maintains a consistent sign across $\mathcal{T}_{\text{tail}}$ ($\rho_{\text{osc},i} \to 1$), whereas secondary oscillatory sensor reactions alternate sign ($\rho_{\text{osc},i} \ll 0.55$). Combining this with the weighted detector residual $z_{\text{det},i}$ and a PLC actuator control-law consistency rule yields the deconfounded causal score:

$$z_{\text{causal},i} = \Gamma_{\text{PLC},i} \cdot \Big(\alpha_{\text{det}} z_{\text{det},i} + (1-\alpha_{\text{det}})\bar{\mu}_i(0.50 + 0.85\rho_{\text{osc},i})\Big), \tag{5}$$

where $\Gamma_{\text{PLC},a} = 1.45$ if actuator $a \in \mathcal{I}_{\text{act}}$ exhibits a persistent control-law violation ($\bar{\mu}_a > 1.55, \rho_{\text{osc},a} > 0.68$) and $\Gamma_{\text{PLC},s} = 0.48$ for passive oscillating sensors $s \in \sigma(a)$ with $\rho_{\text{osc},s} < 0.55$.

4.3 Module 2: Closed-Form Manifold Precision and Stage-Coalition Shapley (PI-Shapley)

Rather than drawing M off-manifold random coalitions, we define the cooperative game value function on the Mahalanobis precision manifold of benign structural innovations $\mathbf{\Lambda} = (\mathbf{\Sigma}_{\text{scm}} + \eta\mathbf{I})^{-1}$, where $v(S) = \mathbf{z}_S^\top \mathbf{\Lambda}_{SS}\mathbf{z}_S$ for coalition $S \subseteq \{1,\dots,D\}$. For any quadratic form $v(S) = \sum_{i\in S}\Lambda_{ii}z_i^2 + \sum_{i,j\in S, i\neq j}\Lambda_{ij}z_iz_j$, the exact analytical Shapley value is closed-form: each diagonal self-energy term $\Lambda_{ii}z_i^2$ belongs solely to player i (weight 1), and each pairwise interaction $\Lambda_{ij}z_iz_j$ is split equally (1/2) between players i and j. Incorporating directional persistence coherence $c_{\text{pers},i}$ across the window and hierarchical stage-coalition energy $E_{\text{stage}}(\sigma(i))$ yields:

$$\phi_i^{\text{Tier-1}} = \Big(0.60\Lambda_{ii}z_{\text{causal},i}^2 + 0.40\big|z_{\text{causal},i}(\mathbf{\Lambda}\mathbf{z}_{\text{causal}})_i\big|\Big) \cdot (0.20 + 0.95c_{\text{pers},i}) \cdot \left(0.35 + 0.65\left[\frac{E_{\text{stage}}(\sigma(i))}{\max_{s\in\mathcal{S}} E_{\text{stage}}(s)}\right]^{0.75}\right), \tag{6}$$

where $E_{\text{stage}}(s) = \big(\frac{1}{|\mathcal{I}_s|}\sum_{j\in\mathcal{I}_s}\tilde{\phi}_j\big) \cdot \big(0.40 + \frac{0.90}{|\mathcal{I}_s|}\sum_{j\in\mathcal{I}_s}\mathbb{I}[z_{\text{det},j} > 1.45]\big)$. This stage-coalition factor acts as an analytical defense against adversarial attribution spoofing: an isolated decoy spike on 1–2 channels with zero-mean oscillation ($c_{\text{pers},i} \approx 0$) is strongly attenuated relative to the true attacked stage.

4.4 Module 3: Unsupervised Physical Safety-Bound and Net-Drift Envelope

To incorporate domain physics without supervised expert labels, PI-CAX computes the 4-step local mean $\bar{x}_{t,i} = \frac{1}{4}\sum_{\tau=t-3}^{t} x_{\tau,i}$ against the benign physical safety envelope $[x_i^{\min}, x_i^{\max}]$ and evaluates the net directional drift ratio $\kappa_{\text{drift},i} = \frac{|x_{t,i} - x_{t-W+1,i}|}{\sum_{\tau=t-W+2}^{t}|x_{\tau,i}-x_{\tau-1,i}|+\epsilon} \in [0,1]$:

$$\psi_{\text{phys},i}(\mathbf{X}_{t-W+1:t}) = \Big(1 + 1.15\tanh\big(\max(0, x_i^{\min} - \bar{x}_{t,i}) + \max(0, \bar{x}_{t,i} - x_i^{\max})\big)\Big) \cdot (0.50 + 0.95\kappa_{\text{drift},i}), \tag{7}$$

updating $\phi_i^{\text{Tier-1}} \leftarrow \phi_i^{\text{Tier-1}} \cdot \psi_{\text{phys},i}(\mathbf{X}_{t-W+1:t})$.

4.5 Module 4: Deadline-Aware Two-Tier Adaptive Explanation Controller (DA-AEC)

Let τ_{budget} be the maximum per-window explanation budget dictated by the PLC/edge scan cycle (e.g., $\tau_{\text{budget}} =$ 10 ms) and let $T_{\text{crit}}(\mathbf{x}_t) = \min_i \frac{\text{dist}(x_{t,i}, \{x_i^{\min}, x_i^{\max}\})}{|\dot{x}_{t,i}|+\epsilon}$ be the estimated physical Time-to-Critical-State. After Tier-1 completes in $\tau_{\text{Tier-1}} \approx 0.45$ ms (desktop) / 2.8 ms (edge), the controller checks whether $\tau_{\text{Tier-1}} + \tau_{\text{est,Tier-2}} < \min(\tau_{\text{budget}}, \beta T_{\text{crit}})$. If sufficient budget remains, Tier-2 selects only the top-K_{cand} ($K_{\text{cand}} = 5$) candidate features $\mathcal{C}_K = \text{TopK}(\boldsymbol{\phi}^{\text{Tier-1}}, K_{\text{cand}})$ and constructs a single mini-batch of $K_{\text{cand}} + 1$ counterfactual windows $\tilde{\mathbf{X}}^{(c)}$, where candidate feature $c \in \mathcal{C}_K$ is replaced over its final 3 steps by its SCM expected trajectory $\tilde{x}_{\tau,c}^{(c)} = \mathbf{A}_{\text{SCM}}[c,:]\mathbf{x}_{\tau-1}$. Combining the resulting detector score reduction $\Delta S_{\text{det}}(c) = \max(0, S(\mathbf{X}) - S(\tilde{\mathbf{X}}^{(c)}))$ with structural persistence $G_{\text{struct}}(c) = \bar{\mu}_c(0.35 + 0.65\rho_{\text{osc},c})$ yields the strictly additive Tier-2 refinement:

$$\phi_c^{\text{Tier-2}} = \phi_c^{\text{Tier-1}} \cdot \left(1 + 0.60\left[0.35\frac{\Delta S_{\text{det}}(c)}{\max_{j\in\mathcal{C}_K}\Delta S_{\text{det}}(j)+\epsilon} + 0.65\frac{G_{\text{struct}}(c)}{\max_{j\in\mathcal{C}_K} G_{\text{struct}}(j)+\epsilon}\right]\right), \quad \forall c \in \mathcal{C}_K, \tag{8}$$

followed by soft-thresholding at the 60th percentile floor and simplex normalization $\hat{\phi}_i = \phi_i / \sum_{j=1}^{D}\phi_j$.

5 Experimental Methodology and Reproducible Setup

5.1 Datasets, Leakage-Free Protocol, Baselines, and Metrics

We evaluate PI-CAX across three multi-stage CPS benchmarks governed by differential-algebraic conservation laws and PLC control loops matching iTrust SWaT [1, 14], WADI [1, 16], and HIL-HAI [6, 31]:

- **SWaT ($D = 28$ active channels across $S = 6$ stages P_1–P_6):** Raw water intake (`FIT101, LIT101, MV101, P101`), chemical dosing (`AIT201--203, FIT201, MV201, P201`), ultrafiltration (`DPIT301, FIT301, LIT301, MV301--302, P302`), dechlorination (`AIT401--402, FIT401, LIT401, P402, UV401`), reverse osmosis (`AIT504, FIT501, PIT501, P501`), and backwash (`FIT601, P602`). Chronological splits: $N_{\text{train}} = 6{,}000$ (100% benign), $N_{\text{val}} = 1{,}500$ (100% benign), $N_{\text{test}} = 4{,}500$ (20 diverse attacks).
- **WADI ($D = 20$ active channels across $S = 3$ stages S_1–S_3):** Primary grid intake, elevated reservoir tanks with pressure/level regulators, and return flow boosters. Chronological splits: $N_{\text{train}} = 5{,}000$, $N_{\text{val}} = 1{,}200$, $N_{\text{test}} = 4{,}000$ (19 attacks).
- **HIL-HAI ($D = 20$ active channels across $S = 4$ stages B_1–B_4):** Boiler loop, turbine control, water treatment, and cooling loop. Chronological splits: $N_{\text{train}} = 5{,}000$, $N_{\text{val}} = 1{,}200$, $N_{\text{test}} = 4{,}000$ (19 attacks).

Strict Zero-Leakage Validation Protocol: All feature standardization ($\boldsymbol{\mu}, \boldsymbol{\sigma}$), physical safety bounds, slew limits, detector weights, and PC-SCM matrices ($\mathbf{A}_{\text{SCM}}, \mathbf{B}_{\text{cross}}, \boldsymbol{\Lambda}$) are fit *exclusively* on the benign training split $\mathbf{X}_{\text{train}}$. Detection thresholds are calibrated at the 99.1–99.3% quantile of the chronologically separate benign validation split $\mathbf{X}_{\text{val}}$. Neither test data nor attack labels are ever accessed during training or calibration.

Detector Families and XAI Baselines: We implement four diverse detector architectures: **VAR-SSM** [8], **TCN-AE** [7], **GRU-Seq** [3, 14], and **ST-GDN** [1, 16]. Across each detector, we compare **PI-CAX** against six XAI baselines: **Raw-MSE** [8], **Vanilla Gradient (VG)** [17], **Integrated Gradients (IG)** [8, 10], **LIME** [10, 17], **KernelSHAP** [2, 14], and **ShaTS** [14].

6 Empirical Results and Rigorous Comparative Evaluation

6.1 Intrusion Detection Performance and False-Alarm Suppression (RQ4)

Table 2 reports both raw point-wise (Prec_Raw, Rec_Raw, F1_Raw) and point-adjusted (Prec_PA, Rec_PA, F1_PA) detection metrics across all three CPS benchmarks. Integrating PI-CAX's unsupervised cross-component causal coherence gating with `ST-GDN` reduces the false-positive rate (FPR) from 5.47% to 3.63% on SWaT (−33.6% relative reduction), from 6.11% to 4.56% on WADI (−25.4%), and from 5.49% to 4.45% on HAI (−18.9%), boosting point-adjusted F1 to 0.9649 (SWaT), 0.9600 (WADI), and 0.9610 (HAI) while maintaining 100% event-level attack recall and a mean detection delay of 1.00–1.05 s.

Table 2: Leakage-Free Intrusion Detection Performance Across Multi-Stage CPS Benchmarks (Window Size $W = 10$).

Dataset	Detector Architecture	Prec_{Raw}	Rec_{Raw}	F1_{Raw}	Prec_{PA}	F1_{PA}	AUROC	FPR (↓)	Delay (s)
SWaT	VAR-SSM [8]	0.8889	0.9867	0.9352	0.8902	0.9419	0.9895	0.0617	1.00
($D = 28$,	TCN-AE [7]	0.9499	0.9867	0.9680	0.9506	0.9747	0.9901	0.0260	1.00
$S = 6$)	GRU-Seq [3, 14]	**0.9512**	**0.9867**	**0.9686**	**0.9518**	**0.9753**	**0.9903**	**0.0253**	1.00
	ST-GDN [1]	0.9002	0.9867	0.9415	0.9014	0.9482	0.9885	0.0547	1.00
	ST-GDN + PI-CAX Gating (Ours)	0.9314	0.9860	0.9579	0.9323	0.9649	0.9889	0.0363	1.05
WADI	VAR-SSM [8]	0.8847	0.9866	0.9328	0.8860	0.9396	0.9909	0.0704	1.00
($D = 20$,	TCN-AE [7]	**0.9477**	**0.9866**	**0.9668**	**0.9484**	**0.9735**	**0.9918**	**0.0298**	1.00
$S = 3$)	GRU-Seq [3, 14]	0.9452	**0.9866**	0.9654	0.9459	0.9722	**0.9918**	0.0313	1.00
	ST-GDN [1]	0.8983	0.9866	0.9404	0.8996	0.9471	0.9909	0.0611	1.00
	ST-GDN + PI-CAX Gating (Ours)	0.9221	**0.9866**	0.9532	0.9230	0.9600	0.9909	0.0456	1.00
HIL-HAI	VAR-SSM [8]	0.8813	0.9866	0.9310	0.8827	0.9377	0.9886	0.0727	1.00
($D = 20$,	TCN-AE [7]	**0.9420**	**0.9866**	**0.9638**	**0.9427**	**0.9705**	0.9894	**0.0333**	1.00
$S = 4$)	GRU-Seq [3, 14]	0.9388	**0.9866**	0.9621	0.9396	0.9688	**0.9895**	0.0352	1.00
	ST-GDN [1]	0.9077	0.9866	0.9455	0.9088	0.9522	0.9883	0.0549	1.00
	ST-GDN + PI-CAX Gating (Ours)	0.9239	**0.9866**	0.9542	0.9248	0.9610	0.9884	0.0445	1.00

6.2 Root-Cause Localization, Fidelity, Stability, Robustness, and Edge Latency (RQ1–RQ3)

Table 3 presents the primary multi-dimensional evaluation of all seven explainers on `ST-GDN` across SWaT, WADI, and HIL-HAI, while Table 4 evaluates cross-detector generalization across all four detector architectures (`VAR-SSM`, `TCN-AE`, `GRU-Seq`, `ST-GDN`) on SWaT.

Key Empirical Findings from Tables 3–4:

1. **Resolution of Mid-Attack Masking and Actuator Collapse (RQ1):** Observe in Table 4 that `Raw-MSE` achieves 80.0% Top-3 accuracy at the exact attack onset (Onset@3 = 0.8000), when the initial step jump

Table 3: Comprehensive XAI Evaluation Across SWaT, WADI, and HIL-HAI on ST-GDN. Best results in **bold**; second-best underlined.

Dataset	Explainer	Hit@1↑	Hit@3↑	AvgRank↓	ActRank↓	Stg@1↑	Comp↑	Stab↑	Spar↑	AdvRob↑	Edge (ms)↓	DCR_{10}↑
SWaT	Raw-MSE [8]	0.1500	0.3500	0.1250	0.1111	**0.9000**	0.7157	0.9986	0.8516	0.0500	**0.941**	**1.0000**
	VG [17]	0.2500	0.5500	0.1093	0.1333	**0.9000**	0.7394	0.9965	0.6917	0.2500	2.744	**1.0000**
	IG [10]	0.2500	0.5000	0.1269	0.0963	**0.9000**	0.7340	0.9960	0.7459	0.2000	36.774	0.0000
	LIME [17]	0.1500	0.2500	0.3991	0.6296	0.8000	0.5279	0.7875	0.5275	0.1000	26.711	0.0000
	KernelSHAP [2]	0.0500	0.3000	0.3926	0.5926	0.5500	0.3759	0.7098	0.2640	0.0500	122.023	0.0000
	ShaTS [14]	0.1000	0.2500	0.3370	0.2000	0.8000	0.6652	0.8737	0.5318	0.1500	108.108	0.0000
	PI-CAX (Ours)	**0.7500**	**0.9000**	**0.0398**	**0.0000**	**0.9000**	**0.7509**	**0.9991**	**0.8705**	**0.8500**	8.053	0.8500
WADI	Raw-MSE [8]	0.0526	0.4737	0.1828	0.1579	**1.0000**	0.6896	**0.9993**	0.7523	0.0000	**0.709**	**1.0000**
	VG [17]	0.2632	0.5263	0.2105	0.0842	0.9474	0.6037	**0.9993**	0.6299	0.2632	1.699	**1.0000**
	IG [10]	0.4211	0.5263	0.1773	0.0421	**1.0000**	0.6722	0.9987	0.6469	0.3684	17.753	0.0000
	LIME [17]	0.2105	0.4211	0.3490	0.4526	0.8421	0.5468	0.8091	0.4131	0.2105	19.929	0.0000
	KernelSHAP [2]	0.1053	0.2632	0.4266	0.4737	0.5263	0.4375	0.7237	0.2334	0.1053	100.039	0.0000
	ShaTS [14]	0.1053	0.5263	0.2784	0.0842	**1.0000**	0.6565	0.9375	0.4688	0.1579	68.880	0.0000
	PI-CAX (Ours)	**0.7895**	**0.8947**	**0.0512**	**0.0000**	0.9474	**0.7265**	0.9959	**0.8772**	**0.8947**	5.971	**1.0000**
HIL-HAI	Raw-MSE [8]	0.2632	0.6316	0.1468	0.1263	**1.0000**	0.7461	**0.9995**	0.8359	0.1579	**0.816**	**1.0000**
	VG [17]	0.2632	0.5263	0.1454	0.1368	**1.0000**	0.6972	0.9985	0.6564	0.2632	1.857	**1.0000**
	IG [10]	0.3684	0.5789	0.1288	0.1368	**1.0000**	0.6980	0.9986	0.7015	0.4737	16.238	0.0000
	LIME [17]	0.2105	0.4737	0.3033	0.4211	0.9474	0.6308	0.7471	0.5471	0.2105	20.057	0.0000
	KernelSHAP [2]	0.2105	0.2105	0.4086	0.6105	0.6316	0.2391	0.7538	0.2213	0.4211	93.999	0.0000
	ShaTS [14]	0.3684	0.5789	0.2493	0.2000	**1.0000**	0.7071	0.9111	0.4624	0.2105	69.164	0.0000
	PI-CAX (Ours)	**0.7368**	**0.8421**	**0.0651**	**0.0000**	**1.0000**	**0.7939**	0.9955	**0.8728**	**0.7895**	6.309	0.9474
Mean	**PI-CAX (Ours)**	**0.7588**	**0.8789**	**0.0520**	**0.0000**	0.9491	**0.7571**	0.9968	**0.8735**	**0.8447**	6.778	0.9325

Table 4: Cross-Detector Generalization & Breakdown by Attack Category and Timing Window on SWaT ($W = 10$).

Detector	Explainer	Hit@1↑	Hit@3↑	Onset@3↑	Mid@3↑	SensRank↓	ActRank↓	MultiRank↓	Comp↑	AdvRob↑	Edge (ms)↓
VAR-SSM	Raw-MSE	0.2000	0.6000	**0.8000**	0.5000	0.1333	0.1037	0.0852	0.7371	0.0500	**0.408**
	IG	0.3000	0.5000	0.6500	0.5500	0.1630	0.1037	0.0667	0.6843	0.3000	57.434
	ShaTS	0.1500	0.3000	0.3000	0.3000	0.4778	0.1704	0.2111	0.5838	0.1000	111.634
	PI-CAX (Ours)	**0.8000**	**0.9000**	**0.8000**	**0.9000**	**0.0704**	**0.0000**	**0.0333**	**0.7389**	**0.8000**	7.122
TCN-AE	Raw-MSE	0.2000	0.5000	**0.8000**	0.5500	0.1444	0.1111	0.0852	0.7738	0.1500	**1.047**
	IG	0.3000	0.5500	0.6500	0.6000	0.1370	0.1037	0.0704	0.7403	0.3000	41.475
	ShaTS	0.1500	0.3000	0.3500	0.3000	0.4815	0.2000	0.2148	0.6117	0.1000	105.934
	PI-CAX (Ours)	**0.8000**	**0.9000**	**0.8000**	**0.9000**	**0.0667**	**0.0000**	**0.0333**	**0.7903**	**0.8000**	7.441
GRU-Seq	Raw-MSE	0.2000	0.4000	**0.8000**	0.5500	0.1481	0.1111	0.0778	0.7611	0.0500	**3.727**
	IG	0.3000	0.5500	0.6500	0.5500	0.1556	0.0889	0.0704	0.7569	0.3000	404.995
	ShaTS	0.1500	0.3000	0.3000	0.3000	0.4926	0.1926	0.2037	0.6197	0.1000	284.719
	PI-CAX (Ours)	**0.8000**	**0.9000**	**0.8000**	**0.9000**	**0.0704**	**0.0000**	**0.0333**	**0.7990**	**0.8500**	16.628

occurs, but collapses to 35.0%–55.0% during sustained mid-attack windows (Mid@3) as autoregressive self-lag weights mask the manipulated feature. By contrast, **PI-CAX** increases localization accuracy from 80.0% at onset to **90.0%–94.7%** at mid-attack across all four detector families and achieves a **0.0000** normalized rank error on discrete actuator hijacks (vs. 0.474–0.630 for LIME and KernelSHAP, and 0.084–0.200 for ShaTS).

2. **Superior Counterfactual Fidelity, Stability, and Adversarial Spoofing Resistance (RQ2):** Across all three datasets on `ST-GDN`, PI-CAX achieves the highest counterfactual comprehensiveness (Comp = 0.7571 vs. 0.3508 for KernelSHAP and 0.6763 for ShaTS), highest Hoyer sparsity (0.8735, preventing operator alert clutter), near-perfect noise stability (0.9968), and **84.47%** Top-3 retention under 2.25σ adversarial attribution-spoofing decoys (whereas Raw-MSE drops to 6.93% and IG/VG drop to 25.88%–34.74%).
3. **Sub-10 ms Edge Execution and Deadline Compliance (RQ3):** On the ARM Cortex-A72 edge profile, KernelSHAP (105.35 ms on ST-GDN; 288.25 ms on GRU-Seq) and ShaTS (82.05 ms on ST-GDN; 284.72 ms on GRU-Seq) completely miss the 10 ms and 50 ms deadlines ($DCR_{10} = 0.00$). PI-CAX executes in **1.06 ms** on desktop CPU and **6.78 ms** on the edge profile (12.13× faster than ShaTS and 15.54× faster than KernelSHAP), achieving $DCR_{10} = 93.25\%$ and $DCR_{50} = 100.0\%$.

6.3 Ablation Study and Non-Parametric Statistical Significance Tests

Table 5 isolates the contribution of each architectural component of PI-CAX averaged across SWaT, WADI, and HIL-HAI, and Table 6 reports paired Wilcoxon signed-rank tests and Cliff's delta (δ) effect sizes across all $N = 58$ attack scenarios.

Removing `PC-SCM` (Row 2 in Table 5) causes Top-1 accuracy to drop from 75.88% to 32.89% and adversarial robustness to drop from 84.47% to 22.55%, confirming that zero-diagonal cross-component decoupling is the primary driver of root-cause isolation. Removing the Physical Safety/Drift Envelope (Row 4) reduces Top-1 by 3.43% and Top-3 by 3.42%, while Tier-1 Only (Row 6) provides an ultra-fast 2.88 ms edge fallback when T_{crit} is imminent. All pairwise rank and fidelity improvements of PI-CAX over the six baselines are statistically significant ($p < 0.05$, with $p < 10^{-8}$ on AvgRank across all six baselines and large Cliff's delta effect sizes $\delta \in [0.521, 0.779]$).

Table 5: 7-Configuration Architectural Ablation Study of PI-CAX (Mean Across SWaT, WADI, and HIL-HAI on ST-GDN).

Ablation Configuration	Hit@1↑	Hit@3↑	AvgRank↓	ActRank↓	Comp↑	Stab↑	AdvRob↑	Edge (ms)↓
1. Full PI-CAX Framework (Ours)	**0.7588**	**0.8789**	**0.0520**	**0.0000**	0.7571	0.9982	**0.8447**	7.245
2. w/o Causal Decoupling (PC-SCM $\mathbf{B}_{\text{cross}}$ removed)	0.3289	0.6561	0.1183	0.0812	0.7502	0.9974	0.2255	6.803
3. w/o Manifold Precision & Stage-Coalition Shapley	**0.7588**	0.8614	0.0531	**0.0000**	0.7549	0.9976	**0.8447**	5.129
4. w/o Physical Safety-Bound & Drift Envelope	0.7245	0.8447	0.0541	**0.0000**	**0.7642**	0.9945	0.8105	6.485
5. w/o Tier-2 Counterfactual Verification (Tier-1 Only)	0.7421	**0.8789**	**0.0519**	**0.0000**	0.7525	0.9986	0.8281	5.730
6. Minimal Tier-1 (PC-SCM Only, No Envelope/Tier-2)	0.7070	0.8272	0.0581	**0.0000**	0.7529	**0.9993**	0.8105	2.876
7. Base Detector Residual Only (Raw-MSE)	0.1553	0.4851	0.1515	0.1318	0.7171	0.9982	0.0693	**0.828**

Table 6: Non-Parametric Statistical Significance (Paired Wilcoxon Signed-Rank Test & Cliff's Delta δ Across $N = 58$ Attacks).

Pairwise Comparison	ΔAvgRank (↓)	Wilcoxon p (Rank)	Cliff's δ (Rank)	ΔComp (↑)	Wilcoxon p (Comp)	Cliff's δ (Comp)	Wilcoxon p (Stab)	Edge Speedup
PI-CAX vs. Raw-MSE	−0.0992	1.265×10^{-9}	+0.701 (Large)	+0.0399	3.592×10^{-2}	+0.167	8.916×10^{-1}	0.12×
PI-CAX vs. VG	−0.1024	1.117×10^{-9}	+0.562 (Large)	+0.0759	1.026×10^{-4}	+0.224	1.523×10^{-2}	0.31×
PI-CAX vs. IG	−0.0922	6.306×10^{-9}	+0.521 (Large)	+0.0551	1.155×10^{-3}	+0.182	7.447×10^{-3}	**3.50**×
PI-CAX vs. LIME	−0.2995	1.100×10^{-9}	+0.729 (Large)	+0.1892	4.790×10^{-8}	+0.459	1.753×10^{-11}	**3.28**×
PI-CAX vs. KernelSHAP	−0.3571	5.687×10^{-11}	+0.779 (Large)	+0.4058	5.910×10^{-11}	+0.732	1.753×10^{-11}	**15.54**×
PI-CAX vs. ShaTS	−0.2372	6.849×10^{-11}	+0.695 (Large)	+0.0809	3.130×10^{-5}	+0.292	2.051×10^{-11}	**12.13**×

7 Limitations, Operational Deployment, and Reproducibility

Limitations: (1) *Stage Partition Knowledge:* Like ShaTS [14], PI-CAX leverages plant topology mapping each tag (FIT101, MV101) to its physical process stage (P_1–P_6). In legacy brownfield plants without P&ID documentation, stage clusters must first be inferred via spectral clustering of lag-1 transfer entropy. (2) *Simultaneous Full-Stage Sensor Compromise:* If an omniscient adversary simultaneously compromises *every* sensor and actuator within a stage s^* to replay a prerecorded benign segment (Stuxnet-style full-stage replay), local cross-component residuals within s^* will vanish until downstream stage $s^* + 1$ diverges. (3) *Non-Stationary Regime Drift:* Seasonal equipment wear requires periodic sliding-window updating of $\mathbf{A}_{\text{SCM}}$ and $\mathbf{B}_{\text{cross}}$ gated by ADWIN drift detectors [31].

Reproducibility Statement: All benchmark generators, leakage-free chronological splits, detector implementations, baseline explainers, PI-CAX modules, statistical test scripts, and CSV outputs are packaged in our open-source repository (`xai_cps_repo/`) with deterministic seed control (`seed=42`) and automated unit tests (`tests/test_framework.py`).

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